



XLSense

Intelligent Spreadsheet Automation Platform

Team Name: Team Code XYZ

Course: CSE299

Section: 15

Instructor Name: Tanzilah Noor Shabnam

Date: April 30, 2026



Group Members

TEAM CODEXYZ

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(Section: 16)

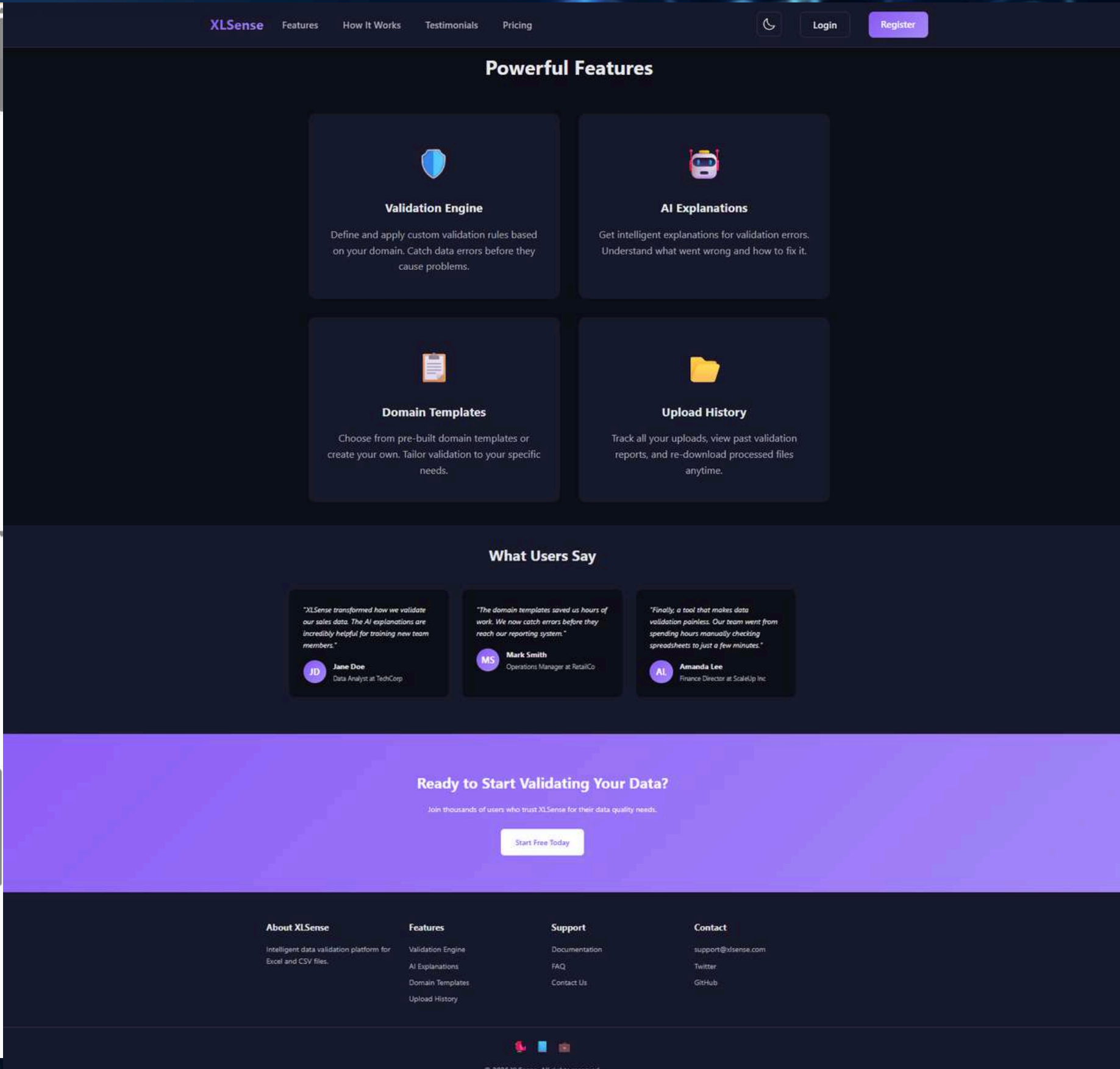
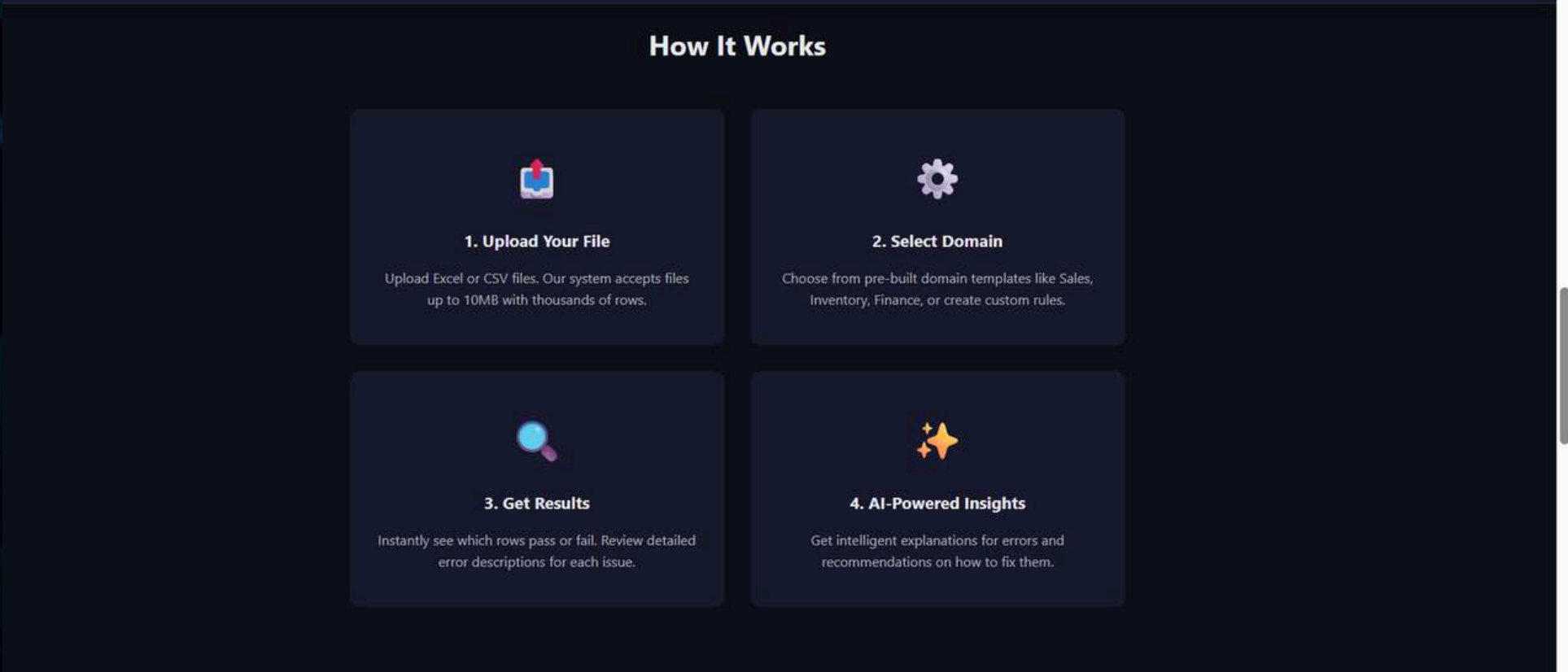
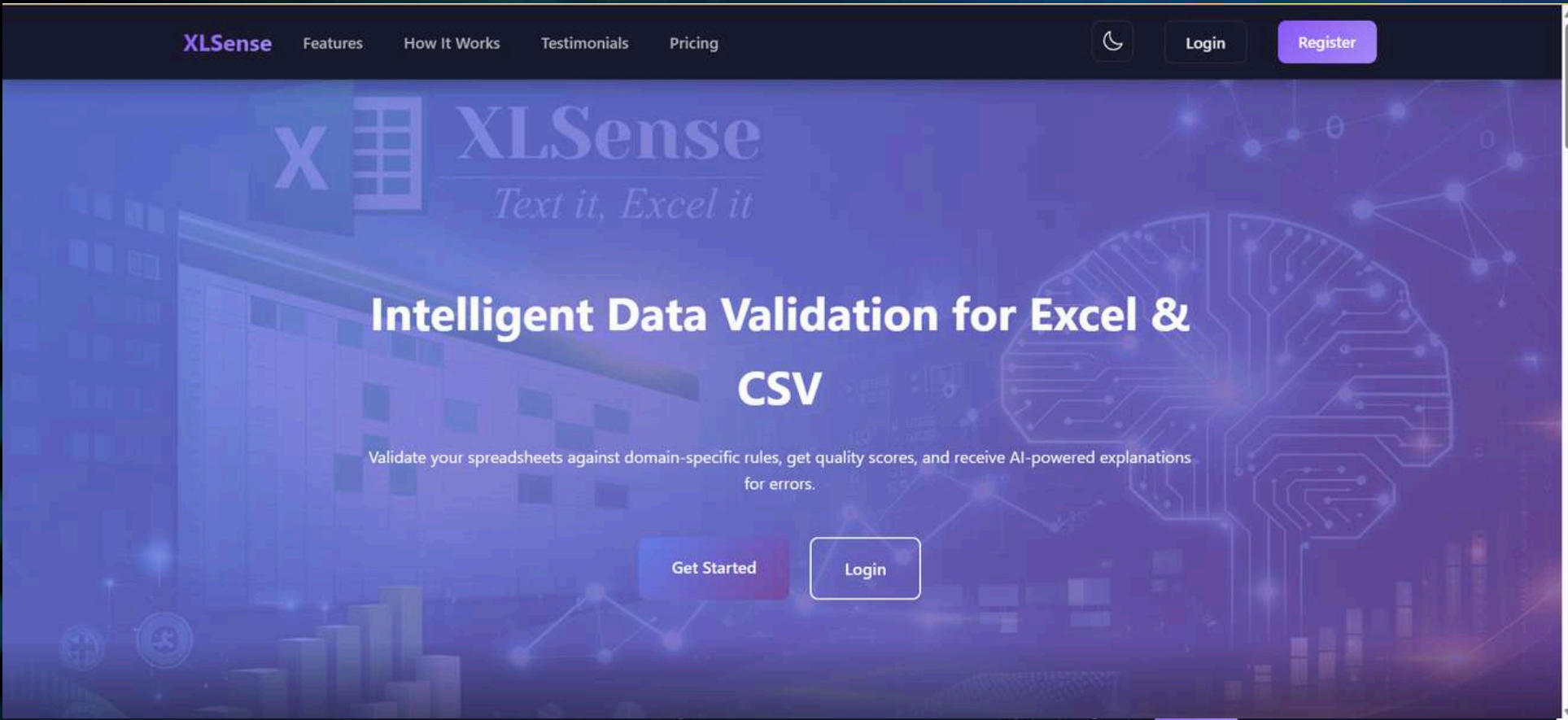


Roadmap

- **Introduction & Motivation**
- **Existing Problems**
- **Proposed Solution (XLSense)**
- **Features & Implementation**
- **System Workflow**
- **Results & Limitations**
- **Future Work**
- **Conclusion & Demo**



HOMEPAGE





Introduction

- Intelligent data workflow & Excel automation platform
- A smart system that handles data logic.
- Turns static table into AI-assisted, dynamic workflows
- Covers multiple domains (office & academic)
- Reduces errors and manual effort





Targeted Audience

- Educational Institutions
- Corporate HR & Finance Departments
- Small & Medium Enterprises (SMEs)
- Administrative & Data Management Teams

Benefits

- Streamlined and automated spreadsheet workflows
 - Reduced operational time and manual dependency
 - Minimized human errors through intelligent validation
 - Faster decision-making with structured data insights
 - Increased productivity across academic and corporate tasks
 - Improved accountability with transparent system logic



EXISTING WORK AVAILABLE

- Microsoft Excel
- Google Sheets

Limitations:

- Manual formula writing
- No validation automation
- No intelligent suggestions



Features

- Automatic column detection & classification
- Domain-specific templates (Academic, HR, Finance, Inventory)
- Intelligent formula recommendation system
- Data validation & error detection
- Audit trail system
- Role-based access control (Admin/User)



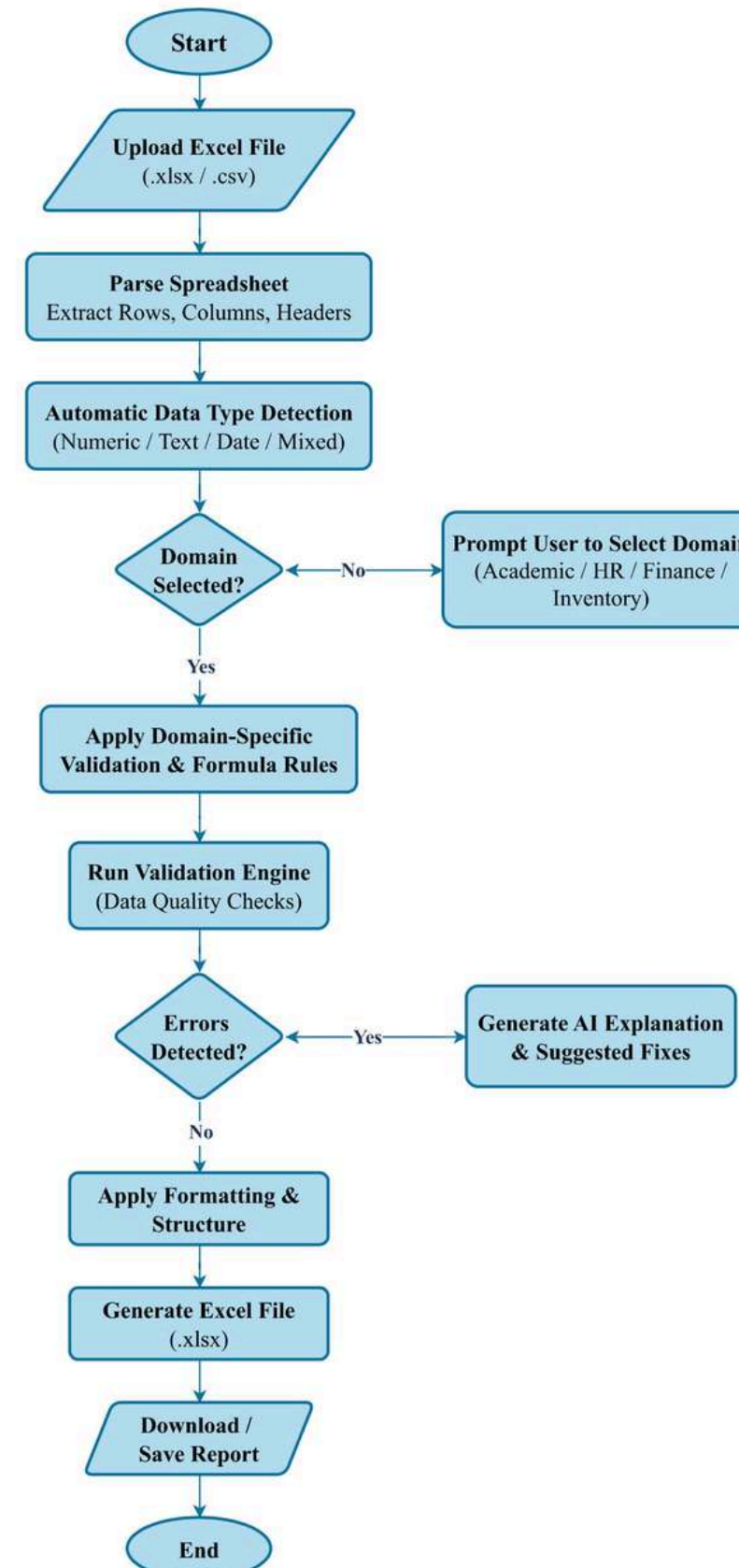
SOLUTION DETAILS

- ➔ **User Authentication**
- ➔ **File Upload & Domain Selection**
- ➔ **Data Parsing & Column Detection**
- ➔ **Data Preview**
- ➔ **Workbook Editor & Column Mapping**
- ➔ **Validation & Processing**
- ➔ **Excel Generation & Download**



Flow Diagram

- User uploads raw data
- System analyzes data patterns
- Data type classification
- Template selection (domain-based)
- Formula & validation application
- Excel file generation
- Feedback & explanation display





IMPLEMENTED FEATURES

User Section

- User login and authentication
- Upload Excel/CSV files
- Select processing domain
- View uploaded file preview
- Track processing status (Uploading / Processing / Completed)
- Use the workbook editor for column mapping
- Download processed Excel file

Admin Section

- Admin login with role-based access
- View all uploaded files (all users)
- Manage domain templates
- Monitor system activity (audit logs)
- Control validation rules and processing logic



Novel Features

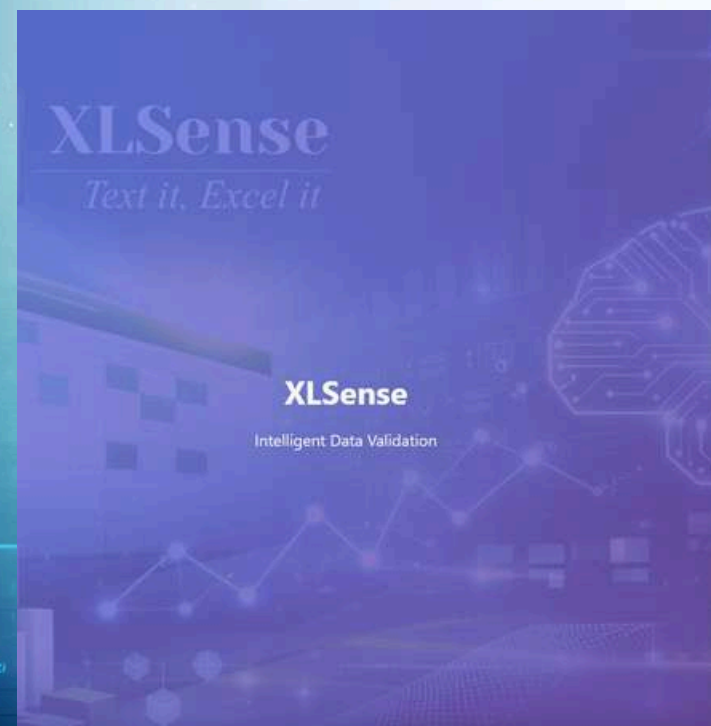
- **Combines Excel automation with AI-based decision support**
- **Provides explainable recommendations**
- **Domain-adaptive spreadsheet generation**
- **Smart validation beyond basic Excel functions**
- **Integrated audit and tracking mechanism**





FUNCTIONAL DETAILS

SIGN UP:



Sign up

Already have an account? [Log in](#)

Username

Choose a username

Required: 150 characters or fewer.

Email

Enter your email

Password

Create a password

Must be at least 8 characters.

Confirm Password

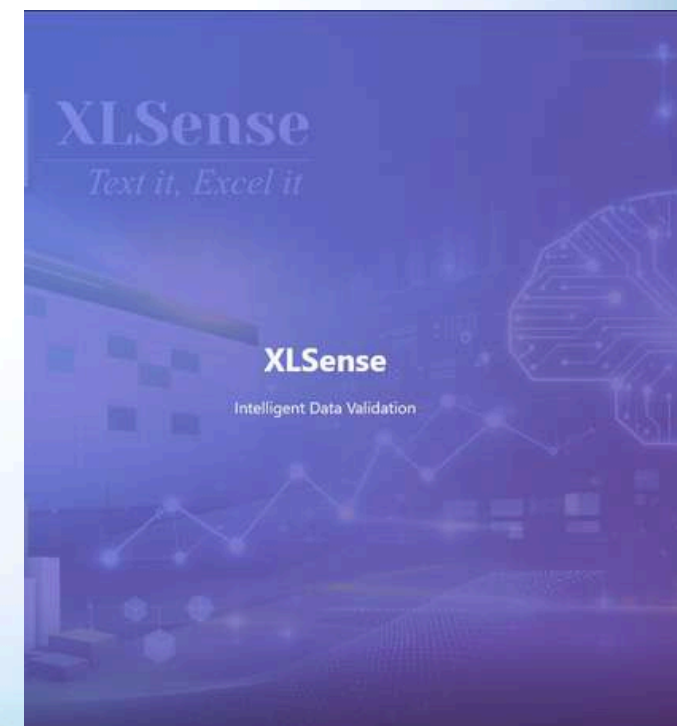
Confirm your password

☐ I agree to the [Terms of Service](#).

Create account

Or continue with

LOGIN:



Log in

Don't have an account? [Sign up](#)

Username

rubaiya

Password

☐ Remember me

[Forgot password?](#)

Log in

Or continue with

G Google

f Facebook



FUNCTIONAL DETAILS

ADMIN DASHBOARD:

XLSense

Dashboard

Audit Logs

Domain Templates

Upload File

Workbook Editor

Upload History

Logout

My Dashboard

Total Files

2

Processed

2

Pending

0

Administrative Control Panel

You have elevated privileges. Use the adjacent tools to monitor system health or manage raw data.

System Stats

DB Admin

File Upload History

Filename	Domain	Status	Upload Time	Action
TA_Excel_Practice_3OzbY5K.xlsx	Academic	Completed	Apr 29, 2026 17:16	Download
TA_Excel_PracticeWrng_RXzbr2j.xlsx	Academic	Completed	Apr 29, 2026 17:16	Download

System Stats

XLSense

Dashboard

Audit Logs

Domain Templates

Upload File

Workbook Editor

Upload History

Logout

Admin Dashboard | XLSense

System Overview

Real-time stats for XLSense validation engine.

TOTAL USERS

7

System Users

TOTAL DOMAINS

4

Active in Database

FILES PROCESSED

3

System Wide

VALIDATION RULES

28

Core Logic Active

Recent System Activity

View All Logs

USER	ACTION TYPE	EVENT DETAILS	TIMELINE
rubaiya	File Save	Saved workbook changes: TA_Excel_Practice_3OzbY5K_edited.xlsx (via editor)	30 minutes ago
rubaiya	File Upload	Successfully uploaded and parsed: TA_Excel_Practice.xlsx Domain: Academic	42 minutes ago
rubaiya	File Upload	Successfully uploaded and parsed: TA_Excel_Practice(Wrng).xlsx Domain: Academic	42 minutes ago



FUNCTIONAL DETAILS

USER DASHBOARD:

XLSense

Dashboard

Domain Templates

Upload File

Workbook Editor

Upload History

Logout

My Dashboard

Total Files

4

Processed

4

Pending

0

File Upload History

Filename	Domain	Status	Upload Time	Action
TA_Excel_Practice_yglUBly.xlsx	Academic	Completed	Apr 29, 2026 18:01	Download
TA_Excel_PracticeWrng_zxCpju6.xlsx	Academic	Completed	Apr 29, 2026 18:01	Download
Inventory_txt_doc_converted.xlsx	Academic	Completed	Apr 29, 2026 18:01	Download
HR_PDF_converted.xlsx	Academic	Completed	Apr 29, 2026 18:01	Download

In Dark Mode:

XLSense

Dashboard

Domain Templates

Upload File

Workbook Editor

Upload History

Logout

My Dashboard

Total Files

4

Processed

4

Pending

0

File Upload History

Filename	Domain	Status	Upload Time	Action
TA_Excel_Practice_yglUBly.xlsx	Academic	Completed	Apr 29, 2026 18:01	Download
TA_Excel_PracticeWrng_zxCpju6.xlsx	Academic	Completed	Apr 29, 2026 18:01	Download
Inventory_txt_doc_converted.xlsx	Academic	Completed	Apr 29, 2026 18:01	Download
HR_PDF_converted.xlsx	Academic	Completed	Apr 29, 2026 18:01	Download



FUNCTIONAL DETAILS

Domain Template

Validation Rules

XLsense

Dashboard

Audit Logs

Domain Templates

Upload File

Workbook Editor

Upload History

Logout

Domain Templates

Library of validation logic and automated formulas.

SELECT DOMAIN

Academic

Inventory

Hr

Finance

Quick search rules or columns...

Validation Rules

RULE NAME	COLUMN	CONDITION
Marks Range Check	Total Marks	$x \geq 0$ and $x \leq 100$
Attendance Percentage Check	Attendance (10%)	$x \geq 0$ and $x \leq 10$
Student ID Check	Student Id	$\text{str}(x).\text{strip}() \neq ""$
Assignment Check	Assignment (10%)	$x \geq 0$ and $x \leq 10$
Grade Check	Grade	$x == \{ "A" \text{ if } \text{Total} \geq 93 \text{ else } "A-" \text{ if } \text{Total} \geq 90 \text{ else } "B+" \text{ if } \text{Total} \geq 87 \text{ else } "B" \text{ if } \text{Total} \geq 83 \text{ else } "B-" \text{ if } \text{Total} \geq 80 \text{ else } "C+" \text{ if } \text{Total} \geq 79 \text{ else } "C" \text{ if } \text{Total} \geq 76 \text{ else } "F" \}$
Mid Check	Mid	$x \geq 0$ and $x \leq 30$
Final Check	Final	$x \geq 0$ and $x \leq 30$

Formula Rules

XLsense

Dashboard

Audit Logs

Domain Templates

Upload File

Workbook Editor

Upload History

Logout

SELECT DOMAIN

Academic

Inventory

Hr

Finance

Valid Enrollment Status

Enrolled

$\text{str}(x).\text{strip}().\text{title}() \text{ in } ["Enrolled", "Dropped", "Registered", "Withdrawn"]$

Attendance Percentage Check

Attendance (5%)

$x \geq 0$ and $x \leq 5$

Assignment Percentage Check

Assignment (5%)

$x \geq 0$ and $x \leq 5$

Status Check

Status

$\text{str}(x).\text{strip}().\text{title}() == ("Dropped" \text{ if } \text{str}(\text{Enrolled}).\text{strip}().\text{title}() == "Dropped" \text{ else } "Withdrawn" \text{ if } \text{str}(\text{Enrolled}).\text{strip}().\text{title}() == "Withdrawn" \text{ else } ("Passed" \text{ if } \text{float}(\text{Total or } 0) \geq 76 \text{ else } "Failed"))$

Formula Rules

FORMULA NAME	TARGET	LOGIC
Total Marks Calculation	Total Marks	Attendance + Assignment + Quiz + Mid + Final
Final Grade Calculation	Grade	$\{ "A" \text{ if } \text{total} \geq 93 \text{ else } "A-" \text{ if } \text{total} \geq 90 \text{ else } "B+" \text{ if } \text{total} \geq 87 \text{ else } "B" \text{ if } \text{total} \geq 83 \text{ else } "B-" \text{ if } \text{total} \geq 80 \text{ else } "C+" \text{ if } \text{total} \geq 79 \text{ else } "C" \text{ if } \text{total} \geq 76 \text{ else } "F" \}$
Final Result	Status	$\{ "Passed" \text{ if } \text{Total} \geq 60 \text{ else } "Failed" \}$



FUNCTIONAL DETAILS

UPLOAD FILE:

XLsense

Dashboard

Domain Templates

Upload File

Workbook Editor

Upload History

Logout

File(s) processed successfully.

Upload Excel or CSV File

Select Domain:

Academic

Select File(s):

Choose File(s)

new 4.csv

Upload File

Latest Uploads

TB_81GB3e7.xlsx

70%

Processing...

XLsense

Dashboard

Domain Templates

Upload File

Workbook Editor

Upload History

Logout

Latest Uploads

new_4_S1RZOy6.csv

TB_81GB3e7.xlsx

new 4.csv

Detected Column Types

Column	Detected Type
name	Text
age	Numeric
marks	Numeric
date	Date

Spreadsheet Preview

name	age	marks	date
Adita	23.0	88.0	10-01-24

Upload File

Workbook Editor

Upload History

Logout

Spreadsheet Preview

name	age	marks	date
Adita	23.0	88.0	10-01-24
Rahim	25.0	76.0	11-01-24
Nila	22.0	91.0	12-01-24
Karim	24.0	84.0	13-01-24
Sara	23.0	79.0	14-01-24
			00-01-00

Excel Preview



FUNCTIONAL DETAILS

Workbook Editor

XLsense

Dashboard

Audit Logs

Domain Templates

Upload File

Workbook Editor

Upload History

Overview

Workbook Editor

Select a file to begin domain mapping and data cleaning.

Original Uploads

Files that haven't been mapped to database fields yet.

FILENAME	DOMAIN	STATUS	UPLOAD TIME	ACTION
TA_Excel_Practice_30zbY5K.xlsx	Academic	Completed	Apr 29, 2026 12:16	Open Editor
TA_Excel_PracticeWrng_RXzbr2j.xlsx	Academic	Completed	Apr 29, 2026 12:16	Open Editor

Edited Workbooks

Files with completed mapping and processing.

FILENAME	DOMAIN	STATUS	LAST MODIFIED	ACTION
TA_Excel_Practice_30zbY5K_edited.xlsx	Academic	Completed	Apr 29, 2026 12:28	View Edited File

uploads/TA_Excel_PracticeWrng_RXzbr2j.xlsx

Undo Redo Add Row Add Column Save & Process

Enter formula (e.g., =SUM(A1:A5))

FX Q1 Formulas

CAI Explain

MID_1	MID_2	MID_1/2(30%)	FINAL_40%	TOTAL	GRADE	STATUS	ENROLLED	BONUS	FINAL GRADE
24	24	35	85.0	B					
23	23	32	82.0	B+					
21	21	30	72.0	C					
18	18	25	61.0	A					
26	26	38	91.5	A+					
22	22	33	78.5	C					
28	28	39	96.5	A					
17	17	24	57.0	C					
20	20	29	69.5	A-					
25	25	36	87.0	B+					
21	21	31	74.0	C					
18	18	26	61.5	C					
27	27	40	97.0	A					
22	22	34	78.0	C					
27	27	38	92.5	A-					

Intelligent Auditor Analysis

Explanation of that validation error:

- The error occurs in a grade calculation formula.
- The formula checks the "total" score and assigns a letter grade based on specific ranges.
- The conditions and corresponding grades are:
 - "A" if total score is 90-92
 - "B+" if total score is 87-89
 - "B" if total score is 83-86
 - "B-" if total score is 80-82
 - "C+" if total score is 79
 - "C" if total score is 76-78
 - "C-" if total score is below 76
- The issue likely arises from inconsistent or incorrect total scores, causing an invalid grade to be assigned.

uploads/TA_Excel_PracticeWrng_RXzbr2j.xlsx

Undo Redo Add Row Add Column Save & Process

Enter formula (e.g., =SUM(A1:A5))

FX C31 Formulas

Intelligent Recommendations

MISSING COLUMN: BONUS

Suggested: Create 'Bonus' column (required for validation).

Formula: None (Manual Data Entry)

MISSING COLUMN: STATUS

Suggested: Create 'Status' column. Use the formula below to calculate it.

Formula: 'Passed' if Total >= 60 else 'Failed'

MISSING COLUMN: ENROLLED

Suggested: Create 'Enrolled' column (required for validation).

Formula: None (Manual Data Entry)

MISSING COLUMN: FINAL GRADE

Suggested: Create 'Final Grade' column (required for validation).

Formula: None (Manual Data Entry)

uploads/TA_Excel_PracticeWrng_RXzbr2j.xlsx

Undo Redo Add Row Add Column Save & Process

Enter formula (e.g., =SUM(A1:A5))

FX Q31 Formulas

Intelligent Auditor Analysis

Explanation of that validation error:

- The error occurs in a grade calculation formula.
- The formula checks the "total" score and assigns a letter grade based on specific ranges.
- The conditions and corresponding grades are:
 - "A" if total score is 90-92
 - "B+" if total score is 87-89
 - "B" if total score is 83-86
 - "B-" if total score is 80-82
 - "C+" if total score is 79
 - "C" if total score is 76-78
 - "C-" if total score is below 76
- The issue likely arises from inconsistent or incorrect total scores, causing an invalid grade to be assigned.



FUNCTIONAL DETAILS

SAVE AND PREVIEW

XLsense Dashboard

127.0.0.1:8000/upload/

Format as Table: Medium

Download Styled Excel

Go Back to Workbook Editor

Style: medium

Sl No	Name	Q1	Q2	Q3	Avg 2 Quiz	In 10%	In 20%	Column1	Assignment	Grade	Final	Assignment (%)	Attendance (%)	Status	Mid	Student Id	Final Grade	Total Marks
1.0	ABDULLAH	10.0	11.0	12.0	11.5	2.875	5.75	0										
2.0	Beli	11.0	12.0	13.0	12.5	3.125	6.25	0										
3.0	Candy	12.0	13.0	14.0	13.5	3.375	6.75	0										
4.0	Dua	13.0	14.0	15.0	14.5	3.625	7.25	0										
5.0		14.0	15.0	16.0	15.5	3.875	7.75	0										
6.0		15.0	16.0	17.0	16.5	4.125	8.25	0										
7.0		16.0	17.0	18.0	17.5	4.375	8.75	0										
8.0		17.0	18.0	19.0	18.5	4.625	9.25	0										
9.0		18.0	19.0	20.0	19.5	4.875	9.75	0										
10.0		19.0	20.0	21.0	20.5	5.125	10.25	0										

Format as Table: Medium

Download Styled Excel

Go Back to Workbook Editor

Style: medium

Sl No	Name	Q1	Q2	Q3	Avg 2 Quiz	In 10%	In 20%	Column1	Assignment	Grade	Final
1.0	ABDULLAH	10.0	11.0	12.0	11.5	2.875	5.75	0			
2.0	Beli	11.0	12.0	13.0	12.5	3.125	6.25	0			
3.0	Candy	12.0	13.0	14.0	13.5	3.375	6.75	0			
4.0	Dua	13.0	14.0	15.0	14.5	3.625	7.25	0			
5.0		14.0	15.0	16.0	15.5	3.875	7.75	0			
6.0		15.0	16.0	17.0	16.5	4.125	8.25	0			
7.0		16.0	17.0	18.0	17.5	4.375	8.75	0			
8.0		17.0	18.0	19.0	18.5	4.625	9.25	0			

DOWNLOADED FILE

TB_81GB3e7_edited_processed (1).xlsx - Excel

File Home Insert Page Layout Formulas Data Review View Acrobat Design Tell me Sign in Share

Clipboard Font Alignment Number Styles

Conditional Formatting

Format as Table

Cell Styles

Cells Editing

A1

sl_no

sl_no	name	q1	q2	q3	avg_2_quiz	in_10%	in_20%	column	assignment
1	ABDULLAH	10	11	12	11.5	2.875	5.75	0	
2	Beli	11	12	13	12.5	3.125	6.25	0	
3	Candy	12	13	14	13.5	3.375	6.75	0	
4	Dua	13	14	15	14.5	3.625	7.25	0	
5		14	15	16	15.5	3.875	7.75	0	
6		15	16	17	16.5	4.125	8.25	0	
7		16	17	18	17.5	4.375	8.75	0	
8		17	18	19	18.5	4.625	9.25	0	
9		18	19	20	19.5	4.875	9.75	0	
10		19	20	21	20.5	5.125	10.25	0	
11		20	21	22	21.5	5.375	10.75	0	
12		21	22	23	22.5	5.625	11.25	0	
13		22	23	24	23.5	5.875	11.75	0	
14		23	24	25	24.5	6.125	12.25	0	
15		24	25	26	25.5	6.375	12.75	0	
16		25	26	27	26.5	6.625	13.25	0	

Sheet



FUNCTIONAL DETAILS

UPLOAD HISTORY:

XLSense

Dashboard

Audit Logs

Domain Templates

Upload File

Workbook Editor

Upload History

Overview

File Upload History

Total Records: 2

Trash

+ Upload New File

USER	FILENAME	DOMAIN	STATUS	DATE & TIME	ACTIONS
rubaiya	TA_Excel_Practice_3OzbY5K.xlsx	Academic	Completed	Apr 29, 2026 17:16	View Report Delete
rubaiya	TA_Excel_PracticeWrng_RXzbr2j.xlsx	Academic	Completed	Apr 29, 2026 17:16	View Report Delete

Validation Report

File: TA_Excel_Practice_3OzbY5K.xlsx

Back to History

DATA QUALITY SCORE

100.00%

This score represents the percentage of rows that passed all validation rules.

✓ Data is Clean!

No errors found in TA_Excel_Practice_3OzbY5K.xlsx. Ready for processing.

Return to History

Intelligent Recommendations

MISSING COLUMN: STATUS

Suggested: Create 'Status' column. Use the formula below to calculate it.

Formula: 'Passed' if Total >= 60 else 'Failed'

MISSING COLUMN: ENROLLED

Suggested: Create 'Enrolled' column (required for validation).

Formula: None (Manual Data Entry)

Validation Report

File: TA_Excel_PracticeWrng_RXzbr2j.xlsx

Back to History

DATA QUALITY SCORE

40.00%

This score represents the percentage of rows that passed all validation rules.

⚠ Issues Detected

Found 19 validation errors that require attention.

ROW	FIELD/COLUMN	ERROR DESCRIPTION	STATUS
2	Attendance (5%)	Attendance cannot exceed 5%.	Failed AI Explain
3	Grade	Manual Entry Error: The Letter Grade in this row does not match the required threshold for the calculated Total marks.	Failed AI Explain
4	Grade	Manual Entry Error: The Letter Grade in this row does not match the required threshold for the calculated Total marks.	Failed AI Explain

✦ INTELLIGENT AUDITOR ANALYSIS

✦ Intelligent Auditor Analysis

Explanation of that validation error:

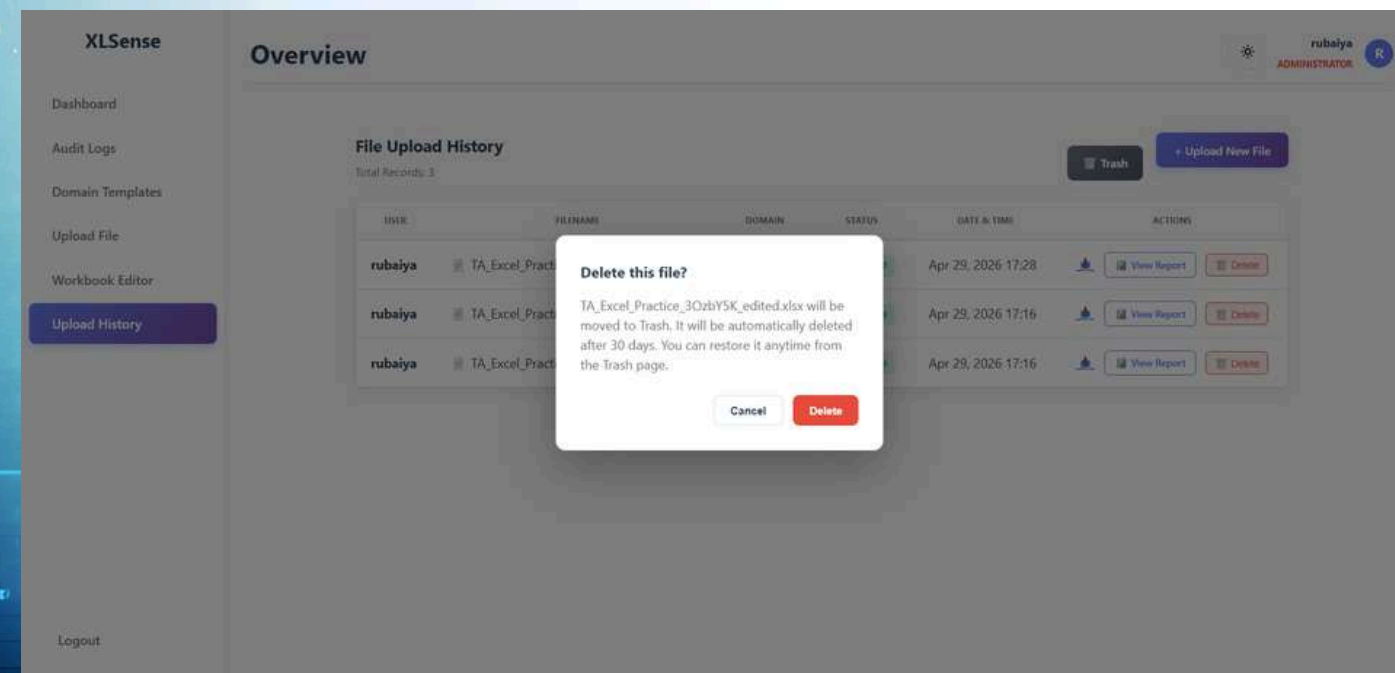
- The validation rule is checking the value in the 'Attendance' column.
- The condition is that the attendance percentage should not be more than 5%.
- The error occurs when the attendance percentage entered exceeds 5%.
- This means the value you entered is more than the allowed 5% attendance, and it needs to be corrected to be 5% or less.



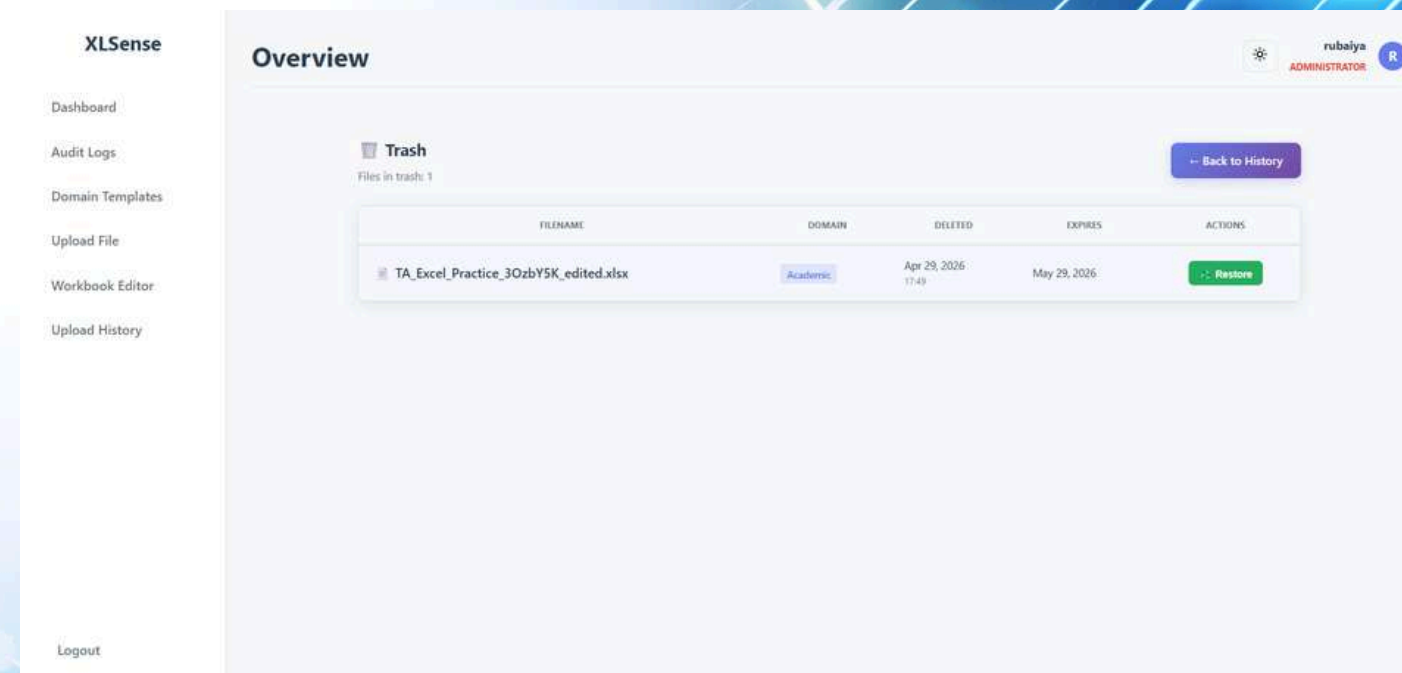
FUNCTIONAL DETAILS

File Delete & Restore

Delete



Restore





FUNCTIONAL DETAILS

Audit Logs

XLsense

Dashboard

Audit Logs

Domain Templates

Upload File

Workbook Editor

Upload History

Logout

Audit Logs | XLsense

rubaiya

ADMINISTRATOR

Audit Logs

3 total records

SEARCH

USER

ACTION TYPE

FROM DATE

TO DATE

Apply Filters

Clear

USER	ACTION	ENTITY	TIMESTAMP	DETAILS / METADATA
rubaiya	File Save	File	Apr 29, 2026 17:28:19	Saved workbook changes: TA_Excel_Practice_3QzbY5K_edited.xlsx (via editor)
rubaiya	File Upload	File	Apr 29, 2026 17:16:27	Successfully uploaded and parsed: TA_Excel_Practice.xlsx Domain: Academic
rubaiya	File Upload	File	Apr 29, 2026 17:16:09	Successfully uploaded and parsed: TA_Excel_Practice(Wrng).xlsx Domain: Academ...



How It's Different From Existing Products

Workbook Editor & Mapping System

- Users can map columns manually
- AI assists in mapping
- Ensures correct validation
- Reduces dependency on raw detection
- System detects columns
- Editor allows user mapping
- Mapping aligns with domain schema
- Then validation & formulas applied



How It's Different From Existing Products

Text/PDF to Excel Conversion

- Supports unstructured data files
- Converts text and PDF into tables
- AI assists in structuring data
- Reduces manual data entry
- Improves usability beyond Excel files
- User uploads .txt or .pdf file
- System extracts raw text
- AI processes and structures data
- Fallback parser ensures reliability
- Output generated as Excel file



METHODOLOGY

- Database design (MySQL)
- Backend processing (Django)
- Data parsing (pandas)
- AI integration (API-based)
- UI development





Software/Hardware Used

Software:

- **Frontend:** HTML, CSS, JavaScript
- **Backend:** Django (Python)
- **Database:** MySQL
- **Data Processing:** Pandas
- **Excel Automation:** openpyxl
- **AI Integration:** API-based NLP service

Hardware:

- **Standard PC/Laptop**
- **Minimum 8GB RAM**
- **Internet connection**
(for API integration)



Not Implemented Features

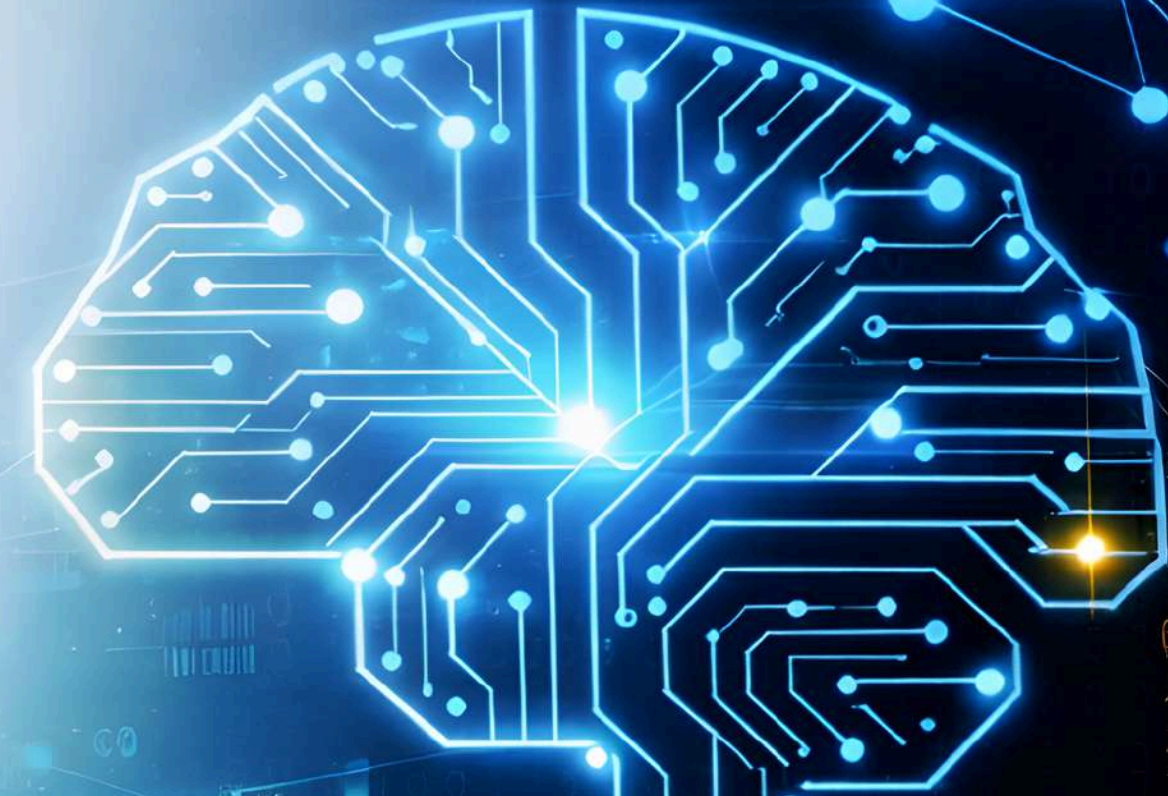
- Fully automated formula engine
- Advanced AI prediction
- Real-time collaboration
- Cloud deployment





LIMITATIONS

- **Requires correct column mapping**
- **AI suggestions are basic**
- **Limited domain templates**
- **Processing not fully automated**





RESULT DETAILS

- **Successful file upload & parsing**
- **Column detection works**
- **Editor allows structured processing**
- **Excel generation functional**





Future Enhancement

- Fully automated processing
- Smarter AI suggestions
- More domains
- Cloud deployment
- Real-time collaboration





What we learnt from project

- Django backend development
- Database integration
- Data processing using pandas
- Team collaboration
- Problem solving



Weekwise Contribution(Labib)





Weekwise Contribution(Adita)

WEEK 1 — Environment Setup & Integration

- Set up Django project and local MySQL database
- Installed required libraries (Django, mysqlclient, pandas, openpyxl)
- Pulled and synchronized project from GitHub
- Analyzed project structure (models, views, templates)
- Tested initial file upload prototype using Django FileField
- Integrated upload system with database model (UploadedFile)
- Ensured compatibility with authentication system

WEEK 5 — Excel Output & Feature Expansion

- Implemented Excel file generation from processed data using pandas & openpyxl
- Developed dynamic download functionality
- Integrated download into “Save and Process” workflow
- Improved data preprocessing (cleaning rows, columns, formatting)
- Fixed multi-file upload issues and improved reliability

WEEK 2 — Upload Module & Processing Workflow

- Implemented secure file upload system (.xlsx, .csv)
- Linked uploaded files with users and domain templates
- Added domain selection before upload
- Implemented file validation and error handling
- Built processing status lifecycle (Processing → Completed / Failed)
- Integrated upload data into user dashboard
- Improved frontend upload UI (multi-file, remove option, progress bar)

WEEK 6 — Editor Integration & Final Workflow

- Connected workbook editor with backend processing
- Implemented full workflow: Upload → Edit → Save → Download
- Ensured preview reflects all user edits (structure, formatting)
- Generated new processed Excel file from edited data
- Ensured downloaded file matches preview exactly
- Added support for Excel formatting (bold, color, etc.)
- Unified complete system data flow
- Fixed bugs and improved overall system stability

WEEK 3 — Parsing & Column Detection

- Built spreadsheet parsing engine using pandas
- Extracted rows, columns, and headers from uploaded files
- Implemented automatic column type detection (numeric, text, date, mixed)
- Added multi-file parsing support
- Displayed structured data preview
- Showed detected column types in UI
- Improved UI stability for large datasets

WEEK 7 — Text/PDF Processing & Conversion

- Implemented extraction for .txt files using Python file read & decode
- Extracted PDF text using PyPDF2 (page-by-page parsing)
- Integrated AI API (requests) to convert raw text into structured CSV format
- Implemented fallback parser (rule-based splitting) when AI fails
- Converted structured data into rows and generated Excel output using openpyxl
- Ensured deleted files are removed from persistent preview (frontend + backend sync)

WEEK 4 — UI Enhancement & System Refinement

- Improved upload interface design and usability
- Ensured domain selection is required before upload
- Implemented persistent preview (data remains after refresh)
- Added latest upload tracking feature
- Implemented real-time upload status feedback
- Added support for .txt and .pdf file uploads
- Refined UI structure and overall system stability



Weekwise Contribution(Rubaiya)

Week 1: Project Setup & Database Design

- Established full project foundation using Django with MySQL backend (via XAMPP)
- Designed and implemented database schema in models.py
- Created core models: DomainTemplate, ValidationRule, FormulaRule, UploadedFile, ValidationResult, AuditLog
- Ran migrations and connected backend with database
- Inserted initial domain templates: Student Grading System, Warehouse Inventory Audit, Employee Payroll & Records, Quarterly Budget Report
- Seeded database with 20+ validation & formula rules
- Built basic project structure: views.py, urls.py, base.html
- Created simple view function to render dashboard
- Developed initial dashboard UI (basic frontend layout)
- Defined domain types

Week 5: Workbook Editor & Domain Mapping UI

- Created workbook_editor.html and workbook_list.html views
- Implemented domain template-based table structure displaying both template columns and uploaded file columns
- Added auto-detection and mapping logic for uploaded files
- Implemented color-coded visual feedback for headers: green (mapped), red (unmapped), blue (template-only missing columns)
- Added rich text formatting tools (font color, bold, italic, underline)
- Built spreadsheet formula/aggregation features (Sum, Average, Count, Min, Max)
- Connected "Save & Process" button to trigger "Excel Preview" and "Download" pipeline, completing core automation workflow

Week 2: Domain Templates & Role-Based Navigation

- Created Domain Template section and UI
- Displayed validation rules and formula rules under each domain template
- Implemented backend logic for dynamic data fetching based on user interaction
- Applied relational data filtering for domain-specific contexts
- Used Django ORM for filtered, domain-based rule retrieval
- Connected frontend UI with backend (views.py → templates)
- Added domain templates to navbar for navigation
- Built reusable layout using base.html (template inheritance)
- Developed routing using urls.py
- Designed and implemented custom Admin Dashboard
- Implemented role-based behavior: Admin → custom dashboard, User → user dashboard
- Added features: System Stats section and DB Admin button (redirect to Django Admin panel)

Week 6: Production Readiness – Formatting, Undo/Redo & Excel Generation

- Redesigned workbook_list.html tables with modern styling and empty-state graphics for new users
- Extended spreadsheet editor with Google Fonts integration: custom font families, sizes, text and background colors per cell
- Enhanced captureState() to track typographic changes, enabling accurate Undo/Redo for all formatting actions
- Added style_data JSON field to UploadedFile model via migration
- Replaced .save() with atomic .update() to avoid overwriting style attributes during file creation
- Implemented dual parser (json.loads + ast.literal_eval) to handle quote discrepancies between Python and JavaScript
- Fixed AttributeError and pandas dtype errors in preprocess_dataframe
- Mapped CSS properties (bold, color, etc.) to openpyxl.styles and used BytesIO + ContentFile to generate downloadable .xlsx files without temporary server storage
- Fixed STATIC_URL / STATICFILES_DIRS to reliably load editor_logic.js and editor_styles.css
- Used print tracing to validate data pipeline from browser to database

Week 3: Validation Engine

- Developed core validation engine with backend logic (utils.py) and frontend interface (report.html) for data quality assessment
- Implemented dynamic rule evaluation using eval() to execute mathematical logic stored in ValidationRule database
- Enabled testing of Excel cell values directly against database-defined rules
- Expanded validation rule set to improve diagnostic coverage
- Defined relational links between validation rules and uploaded files

Week 4: Data Quality Scoring & Formula Recommendations

- Implemented intelligent formula recommendations to detect missing columns and suggest automated data-entry logic
- Integrated data quality scoring system generating real-time percentage score (valid rows / total rows)
- Used eval() in utils.py to dynamically execute Python-style logic from database against spreadsheet data
- Leveraged Pandas iterrows() and pd.isna() to sanitize NaN values for stable numerical processing
- Applied string-based comparison logic (str(val).strip()) to detect manual entry errors in non-numerical fields
- Optimized find_best_column utility using alphanumeric mapping to link database rules to user columns despite inconsistent naming or spacing

Week 7: Advanced Formulas, Batch Formatting & AI Feedback

- Expanded function library: added IF, COUNTIF, COUNTA, XLOOKUP, and VLOOKUP
- Implemented smart range detection – automatically identifies vertical/horizontal contiguous cells for faster formula application
- Added batch formatting: click-and-drag multiple cell selection to apply font style, size, text color, and fill color to all selected cells
- Streamlined formula interface – consolidated individual formula buttons into a single "Formulas" menu panel
- Displays formula syntax and examples when interacting with the formula menu
- Fixed active-cell focus issue, preventing unwanted page jumps after applying formulas
- Improved grade validation: fixed existing grade column rules and added Curved Bonus, Final Grade, and Enrollment Status validation
- Resolved edge-case bugs when columns are missing during validation
- Enhanced AI feedback explanation prompt to generate more accurate, user-specific feedback on grid errors
- Eliminated duplicate validation errors (old results deleted before new ones are created)
- Improved real-time cell validation for columns mapped to database names (e.g., Quiz, Mid, Final)



Conclusion

- **XLSense automates complex Excel workflows**
- **Reduces manual effort and human errors**
- **Enhances productivity in academic & office tasks**
- **Provides intelligent and explainable spreadsheet solutions**





Thank You 😊





PROJECT DEMO





Questions?

